Limkokwing Library Management System

Module: PROG315 – Object-Oriented Programming 2
Assignment: Basic API Structure with Open-Software
Semester: 04 | March 2026 – July 2026
Institution: Limkokwing University of Creative Technology, Sierra Leone

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Description

A digital library management system implemented in Python using both synchronous and asynchronous programming. The system allows students and staff to search for books, borrow and return them, and track overdue loans and fines — all without requiring physical presence at the library.

Multiple users can borrow or return books at the same time using Python's `asyncio` module and `asyncio.gather()`, which runs concurrent requests without blocking.

Features

- Search the library catalogue by title, author, or category
- Borrow available books with an automatic 14-day loan period
- Return books with automatic overdue fine calculation (Le 5,000 per day)
- View all active overdue loans with accumulated fines
- Concurrent multi-user support via `asyncio.gather()`
- Full type annotations throughout (PEP 484)
- No external packages required — uses Python standard library only

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Operations

Function	Operation	Description

`sync_search_books()`	GET /books	Search the catalogue
`sync_borrow_book()`	POST /borrow	Borrow an available book
`sync_return_book()`	PUT /return	Return a book and calculate fine
`sync_overdue_loans()`	GET /overdue	List all overdue loans and fines

Project Structure

```
limkokwing-library/
├── main.py          # Complete source code
├── README.md        # Project documentation
└── .gitignore       # Files excluded from version control
```

How to Run

No external packages are required. Python 3.10 or higher is recommended.

```
# Clone the repository
git clone https://github.com/<your-username>/limkokwing-library.git
cd limkokwing-library

# Run the simulation
python3 main.py
```

The script will run four demonstrations in sequence:

1. **SYNC** — searches the catalogue by category
2. **SYNC** — checks for overdue loans
3. **ASYNC** — three users borrow books simultaneously
4. **ASYNC** — two users return books simultaneously

Expected Output

```
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```

```
  Limkokwing Library Management System
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  SYNC  |  GET /books  -->  category=Programming
==============================================================
  Status  : success
  Results : 1 book(s) found
    [B002]  Python Crash Course  |  Eric Matthes  |  Available

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  ASYNC  |  POST /borrow  -->  3 users simultaneously
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  [U010]  --> Requesting book B001 ...
  [U011]  --> Requesting book B002 ...
  [U012]  --> Requesting book B004 ...

  [U010]  SUCCESS - Borrowed "Clean Code"  |  Loan: L002
  [U011]  SUCCESS - Borrowed "Python Crash Course"  |  Loan: L003
  [U012]  SUCCESS - Borrowed "Artificial Intelligence"  |  Loan: L004

  All 3 requests completed in 0.151s  (concurrent)
```

SDG Alignment

This project aligns with **SDG 4 – Quality Education**.

By digitising the library's borrowing process, the system enables equitable access to academic resources for all Limkokwing students regardless of their physical location or daily schedule. Students who cannot visit the library in person can still search the catalogue, check availability, and manage their loans remotely from any internet-connected device.

Academic Integrity

This project was completed in accordance with Limkokwing University's policies on academic integrity and plagiarism.

Collaborator Access

Lecturer: amandus.bcoker@limkokwing.edu.sl

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